

# Online Learning Platform: Internship Roadmap

Core Task Distribution & 2-Week Implementation Plan

## 1. Core Task Distribution

| FRONTEND (UI/UX)                                                                                                                                                                                                                                                       | BACKEND (LOGIC)                                                                                                                                                                                                                                                                            | DATABASE (SCHEMA)                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Register/Login/Forgot Pwd</li><li>• Course Listing &amp; Details</li><li>• Student Dashboard</li><li>• Admin Basic UI</li><li>• Responsive Layout</li><li>• Form Validations</li><li>• API Integration (Axios/Fetch)</li></ul> | <ul style="list-style-type: none"><li>• Auth APIs (JWT/Sessions)</li><li>• Course &amp; Enrollment CRUD</li><li>• Progress Tracking Logic</li><li>• Search &amp; Filter APIs</li><li>• Role-Based Access (RBAC)</li><li>• Admin Approval Flow</li><li>• Logging &amp; Validation</li></ul> | <ul style="list-style-type: none"><li>• Users &amp; Roles Tables</li><li>• Courses &amp; Lessons Tables</li><li>• Enrollments &amp; Progress</li><li>• Password Reset Tokens</li><li>• Audit/Log Tables</li><li>• Foreign Key Constraints</li><li>• Indexing Strategy</li></ul> |

## 2. Implementation Schedule (2 Weeks)

| Day   | Frontend Tasks                                                 | Backend & Database Tasks                                          |
|-------|----------------------------------------------------------------|-------------------------------------------------------------------|
| 1 - 3 | Static UI for Auth and Course Catalog. Setup Router.           | DB Table creation. Setup Auth APIs (Registration/Login logic).    |
| 4 - 6 | Connect Login/Reg to API. Build Course Detail & Enrollment UI. | Course CRUD APIs. Enrollment Logic. Secure routes with JWT.       |
| 7 - 9 | Student Dashboard & Progress Bars. Admin UI components.        | Progress Tracking API. Search/Filter logic. Admin approval flows. |
| 10    | Final Polish, Error States & Bug Fixing.                       | Performance indexing, Final logging, API Documentation.           |



### Critical Items Missing from Original List

To make the platform truly functional, ensure these "Bridge" tasks are assigned:

- **State Management:** How the frontend keeps the user logged in between pages (e.g., Redux/Context).
- **Media Storage:** Logic to store and stream video lessons/PDFs (AWS S3 or Cloudinary).
- **Password Security:** Backend must hash passwords (bcrypt) before DB storage.
- **Cross-Origin Resource Sharing (CORS):** Backend configuration to allow the Frontend to communicate with it.

- **File Upload APIs:** Backend logic to handle multi-part form data for course assets.